

Developement of Computational Model for Estimating Marine Pollution in Ports

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Abstract

The aggressively growing problem of marine pollution and environmental degradation calls for urgent action taken by researchers into tackling said issue. The measures for environmental protection and monitoring are stricter than ever, carving the need for continuous data streams that are assessed in quality and properly handled to fully inform the decision maker. Within the scope of this research falls the processing of biochemical data collected from various sources in the Adriatic region, and the extraction of critical information regarding trends, ranges and models used in the development of an early warning system against marine pollution.

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1 Introduction

Before taking a deep dive into the specifics of a very complicated topic, one must take the time to gain a better understanding of the different factors and components that comprise sea water pollution. While it may only seem like a rough outline, a definition is the most comprehensible way to start off a more rigorous study. The United Nations definition of marine pollution is as follows:

*Direct or indirect introduction by humans of **substances** or **energy** into the marine environment (including estuaries), resulting in harm to living resources, hazards to human health, hindrances to marine activities including fishing, impairment of the quality of sea water and reduction of amenities.*

UN Definition [21]

Said undesirable introduction of matter into oceans, is at the center of this study. The aim is to identify different types of pollution and understand the ways in which it is introduced to the environment. Using that knowledge the issue can be reduced to a finite set of measurable parameters that describe the phenomena, thus thorough monitoring and modeling can potentially predict critical conditions for pollution.

1.1 Pollution Types

As the definition suggests, the most basic categorizations of sea water contamination, is *substances* and *energy*, both of which are used as umbrella terms, covering a range of different types of pollutants. Further expanding the term substances, it describes any kind of matter discarded into oceans due to human activities. Said matter can be easily distinguishable litter, or chemicals, be it nutrients in excess or toxins.

Marine Debris Any manufactured or processed solid material that is disposed of in the marine or coastal ecosystem can be considered debris pollution[1]. Out of the five main categories (plastic, paper, metal, textile, glass and rubber), plastic is the most destructive. Ranging in size from regular product packaging such as bottles, down to microplastics, it impacts marine organisms, their environment, and in extent humans. It is estimated that there are 51 trillion particles of microplastics, caused from the eventual breakdown of plastic waste, floating on the global ocean surface. It is evident that when ingested by small organisms, they get absorbed into their tissues, leading to the migration of microplastics up the food chain, eventually reaching human consumers[40].

Nutrients The excess of chemicals that are otherwise valuable to aquatic life, results in an increase of primary productivity (autotrophs, mainly plants and algae). This nutrient enrichment process is called eutrophication, which leads to algal blooms that can support the increase of invertebrates and fish. The inevitable bacterial degradation of the excess biomass deplete the water body's oxygen creating states of hypoxia[32]. Most common nutrients responsible, are nitrogen and phosphorus, both being crucial to life preservation. Nitrogen plays an important role in protein production, and is usually found in forms such as nitrates (NO_3^-), nitrites (NO_2^-), and ammonium (NH_4^+) while phosphorus is found in nucleic acids (RNA, DNA), energy molecules such as ATP and generally as phosphates (PO_4^{3-})[41]. Due to their properties, both nutrients are heavily used in agriculture in the form of fertilizers, while other sources are burning of fossil fuels and sewage treatment plants[38].

Toxins Tiny particles of toxic chemicals such as heavy metals, when taken up by marine organisms, in a process called bioaccumulation, lead up to upward concentration in food chains. Examples of persistent toxins include TBT(anti-fouling paints), DDT(insecticide), PCBs(carcinogenic chlorine compounds used in production of carbonless copy paper), generally pesticides and radioactive waste. Heavy metals (such as mercury, lead, nickel, etc) are also excellent examples of marine waste finding its way into human food products raising the fear of poisoning[6].

Also falling under chemical pollution, the abundance of carbon dioxide (CO_2) in earth's atmosphere due to human activity, when combined with water creates carbonic acid (H_2CO_3), a weak acid that dissociates into hydrogen anions. By this mechanism, sea water's pH is lowered to more acidic values, creating the phenomenon known as "ocean acidification"[29]. On the other hand, energy is more of an indirect form of contamination, referring to types of energy input to the ocean ecosystem, that result in the altering of its physical properties¹.

Light The presence of artificial light, especially during nighttime, disrupts the natural circadian rhythms of marine animals. Predator species are able to hunt prey easily, reproductive cycles can be disturbed, and sea turtle hatchlings using light cues to navigate become disoriented.

Sound It's worth noting that sound travels more rapidly and over larger distances in water than it does in our atmosphere. For that very reason, noise pollution is a greater threat to aquatic biota, confusing animals which rely on sonar signals to feed, mate and navigate(e.g. cetaceans), forcing them to change behavior[8]. An extreme example is the acoustic trauma causing injuries in the statocyst sensory epithelium of Cnidarians (specifically Mediterranean Scyphozoan medusa)[34]. Common sources of anthropogenic noise pollution are passing ships, oil exploration and naval low-frequency active sonar.

¹Notice the difference between physical and chemical properties. Chemical composition of water refers to parameters such as pH, dissolved oxygen etc. Physical properties are linked to characteristics like temperature, turbidity, color, taste and odor.



Figure 1: Mass death of fish due to golden algae toxic bloom in US[13].

Heat Thermal pollution refers to the temperature changes of sea water due to external factors, such as human interference. The discharge of hot or cold water from power plants and factories (e.g. using water as coolant) cause abrupt changes of ambient temperature, known as "thermal shock". Elevated temperatures can lead to increase of metabolic rates causing malnutrition due to limited resources. Depletion of dissolved oxygen due to reduced solubility in warmer liquids can also have adverse effects on marine life[9].

It's worth noting that, while not previously mentioned, biological pollution as in the introduction of invasive aquatic biota (for example through the ballast discharge of a cargo ship) is also a concerning form of fouling[30].

1.2 Pollution Pathways

For accurately modeling the mechanics of pollution, it is important to examine how pollution reaches the marine environment. There are sources of contaminants, and pathways that lead to the polluted ecosystems. Sources can be classified as *point*(PS) or *non-point source pollution*(NPS), while types of input can be direct discharge, land runoff and atmospheric pollution. Point and non-point simply refer to the spatial identifiability of the contamination: A factory discharging waste to a specific point in a river or ocean is easier to model compared to ill-defined agricultural runoff and wind-blown debris(non-point)[12].

Direct Discharge Direct entering of pollutants into water bodies from sewage and industrial waste, outflow of acidic water from metal or coal mines. The shipping industry is causing many forms of direct pollution, such as oil spills, engine oil disposal, discharging of ballast water, toxic paints, even carrying harmful algae and invasive species[9][6].

Land Runoff Farming and urban activities enrich soil in particles containing carbon, nitrogen, phosphorus and minerals, resulting in a surplus of nutrients entering water[38]. Stormwater originating from roads and driveways introduces petroleum and heavy metals to any body of water it gets absorbed into.

Atmospheric Windblown dust and debris, or high concentrations of carbon dioxide in the atmosphere eventually come into contact with sea water [29].



Figure 2: The destruction caused in Lago Agrio by Texaco(now owned by Chevron Corporation) and the state owned Petroecuador. They are responsible for the disposal of toxic petroleum waste, and hundreds of crude oil leaks in the tropical forest of Ecuador[10][18].

1.3 Importance of an Early Warning System

By now it should be obvious that almost everything humans do, from the basic growing of food to the complex manufacturing of technological marvels, releases pollution into the environment. That makes it impossible to abruptly stop all polluting activities, since the gradual leaning towards cleaner energy, and pollution free manufacturing takes time. This very issue marks the importance of being able to predict critical conditions, and taking mitigation action on time. By introducing the term "**decision maker**", an early warning system is able to inform such an individual of a water body's exact chemical state, and even predict (based on previously gathered data) the risk of pollution. Working much like a **Decision Support System** (DSS), the decision maker is able to make informed choices to avoid undesirable outcomes.

2 Modelling and Quantifying

Careful studying and monitoring of marine environment with the goal of extracting parameters capable of describing state of contamination, has been attempted numerous times, from various countries. The findings are usually published as guidelines, used in assessing water quality in a specific geographic region, but the normal ranges they report can be used as a starting point. Used within this study, are guidelines available to the public such as *USEPA 1992*(United States)[39], *CCME 1999*(Canada)[23], *ANZECC 1992*(Australia & New Zealand)[4] and *ANZECC/ARMCANZ 2000*(Australia & New Zealand)[5], and most importantly (and most recent), UNEP/MAP publications and their results[22][20]. The parameters used are named **stressors**, for the effect they have on the environment when their values are abnormal. Stressors are physical, chemical or biological factors that cause an adverse effect on aquatic ecosystems[7]. The proposed threshold values and ranges, are used to determine an ecosystem's quality, indicating safe conditions and alerting in case of degradation.

This study aims to combine guideline thresholds with data acquired from sensors located on the Adriatic shoreline. While there are many programs in the European area working on monitoring ocean water state, the sensors used are placed, maintained and operated by the European program **SEAVIEWS**(**SE**ctor **A**daptive **VI**rtual **E**arly **W**arning **S**ystem for Marine Pollution), lead by the National Technical University of Athens[17][16].

2.1 Parameter & Geospatial Information

The map shown in figure 3 describes the spatial dimension of the data used within this paper. Each sensor is capable of measuring 9 physicochemical stressors, namely:

- Temperature ($^{\circ}C$)
- Salinity (*ppt*, or *PSU*)
- Conductivity ($\frac{mS}{cm}$)
- Dissolved Oxygen ($\frac{mg}{L}$)
- Dissolved Oxygen Percentage (%)
- pH
- Oxidation Reduction Potential (mV)
- Total Dissolved Solids ($\frac{g}{L}$)
- Turbidity (*NTU*)

To understand each parameter, the factors that affect its stability, and the relationships between stressors, it is important to examine each separately.

Temperature Along with salinity, they are the most valuable and often used parameters in sea water monitoring. There are even specialized autonomous gliders used for this exact reason[36]. It is a **physical** parameter, affecting organisms by increase or decrease. Aquatic ecosystem functioning is very closely regulated by temperature: Excess heat or cold are considered to be forms of thermal pollution. Speed of sound in water, thermal and electrical conductivity all increase in magnitude as temperature increases. Solubility of gases though, such as carbon dioxide, nitrogen, hydrogen and oxygen all decrease with rising temperatures[5][27][41]. Abbreviation used: **TMP**

Salinity, Conductivity & Total Dissolved Solids All three measurements are representing roughly the same thing: The total concentration of inorganic ions (salts) in the water [5]. Salinity is usually measured in PSU (Practical Salinity Units), equivalent to ‰ concentration. Oceans have a salinity of around 35‰, while the in the Adriatic region a range of 35-40‰ is found[33]. It is important to note that the concentration of salts is heavily influenced by the ecosystem studied. For example, coastal sites, affected by freshwater inputs will appear to have lower salinity. Lower salinities allow for higher oxygen concentration and coupled with the temperature effects, warm hypersaline areas maintain some of the lowest oxygen saturation values[26]. Abbreviations used: **SAL, CND, TDS**

pH Describing the concentration of hydrogen ion as a negative logarithm, a unit change in pH, results in a tenfold change in hydrogen ion concentration. A rough range for sea water is between 7 and 9[5][20], while changes occur in different depths and at different times of the day. Solar radiation, while promoting photosynthesis and warming the sea surface, decreases the amount of carbonic acid raising the pH. By the same mechanism, in deeper waters pH decreases. Note that due to the buffering capacity of the marine environment, pH tends to have a stable value whether it refers to oceans, or estuarine waters[25]. Acidification is detrimental to aquatic biota since it affects toxicity, shell thickness and hardness, photosynthetic abilities, and even inhibits cell aggregation of embryos. Abbreviation used: **PH**

Oxygen Concentration & Saturation Much like in humans, marine animals need oxygen to survive. The transfer of oxygen is only efficient above certain concentrations. During photosynthesis, oxygen is produced, and it is consumed during respiration and decomposition. These processes have a diurnal variation, since photosynthetic processes generally require sunlight. Oxygen saturation is derived from the percentage of dissolved oxygen relative to that when completely saturated at the temperature of the measurement site[4]. The solubility of oxygen in water is inversely correlated with temperature and salinity[24], while decrease leads to elevated toxicity of certain chemicals. Proposed limits vary from 3 $\frac{mg}{L}$ to 8 $\frac{mg}{L}$ [5][7]. Abbreviation used: **DO, DOP**



Figure 3: Sensing sites locations in the Adriatic Sea.

Oxidation Reduction Potential Closely linked to the capacity of a water body to self-cleanse, it is affected by depth, pH and dissolved oxygen (and other oxidants). The oxido-reduction potential can monitor water quality by reporting the disinfection potential, since a high value implies abundance of oxidants meaning more efficient breaking down and decomposing of dead tissue. In general ORP decreases with increasing pH, and increases rapidly with increasing oxidant dosage[41]. Proposed limits are between 300 and 500 mV . Abbreviation used: **ORP**

Turbidity A visual property describing the cloudiness of a body of water, caused by suspended or dissolved particles[28]. The unit of measurement is called *NTU* (Nephelometric Turbidity Unit). The higher the *NTU*, the more turbid the water is, appearing less clear. Anything under 20 *NTU* usually appears clear, and over 500 is no longer transparent. UNEP/MAP applies a max value of 150 *NTU* [22][20]. Abbreviations used: **NTU**

2.2 Dataset

The data used in this study are a **timeseries** of these 9 values. The locations are Greek ports on the Adriatic/Ionian sea, specifically Patras, Aigio, Actium and Preveza. Due to difficulty with the datasets (specifically missing values in dissolved oxygen percentage, DOP and salinity, SAL) some values were not used. It is assumed that conductivity is a good approximation of salinity with a simple linear transform, while dissolved oxygen and temperature are enough to describe oxidant state. A rough look on the size of the dataset is shown in table 2.

Indicator	TMP	SAL	PH	DO	ORP	NTU
Type	Physical	Physical	Chemical	Chemical	Chemical	Physical
Stressor	Increase or Decrease	Increase or Decrease	Increase or Decrease	Decrease	Decrease	Increase
Correlations	Inv.: DO Prop.: CND, PH	Inv.: DO Prop.:	Inv.: ORP, DO	Inv.: PH, TMP, CND Prop.: ORP	Inv.: PH Prop.: DO	
Range	10-30	35-40	7-9	4-20	...-600	
Guide Low	<5	10	6	3 or 20% saturation	200	0
Guide High	35-40	40	10	-	400	150

Table 1: Overview of parameter values and correlations

	Count	Start Date	End Date
aktio	106159	2021-09-09 10:27:00	2022-09-30 23:55:42
preveza	106591	2021-09-08 11:40:00	2022-09-30 23:55:42
aigio	100798	2021-07-08 13:31:00	2022-09-30 23:55:42
patra	101145	2021-07-30 00:00:00	2022-09-30 23:55:42

Table 2: Review of the available dataset

3 Anomaly & Outlier Detection

As with any sensor measurement, errors are bound to appear. A first glance at the data will uncover the most obvious outliers with help from table 1. The example in table 3 shows that there are invalid values, such as $\text{pH} > 14$ and 0 concentration of oxygen.

In figure 5 it is unfortunately probable that there is a malfunctioning sensor. Other than that, the dataset is normal. With the same approach, Patras is showing similar anomalous behaviour in figure 6 which makes it impossible to work without removing the invalid values first. For that, an anomaly detection toolkit is used, namely ADTK. The very same toolkit is used to identify anomalous values and warn about the possibility of pollution. For the reasons described above, while counter intuitive, outlier detection will precede any exploratory data analysis.

3.1 Anomaly Detection Algorithms

The packages used throughout the study are ADTK[3], scikit-learn[31] and sktime[19]. In all 3 packages, the same mechanism is used: Model selection $\Rightarrow$ `.fit()` data on the model $\Rightarrow$ `.detect()` for outliers and `.predict()` for forecasting. Note that timeseries data can be handled as univariate, or multivariate, with the latter proving useful when the independent variables are correlated. Algorithms used are discussed in more detail:

Univariate

`adtk.detector.PersistAD`

Compares time series values with the values of their preceding time windows, and identifies a time point as anomalous if the change of value from its preceding average or median is anomalously large. **Parameters:** `c`, factor determining the bound of normal range. **side**, whether to detect positive, negative or both anomalous changes.

`adtk.detector.GeneralizedESDTestAD`

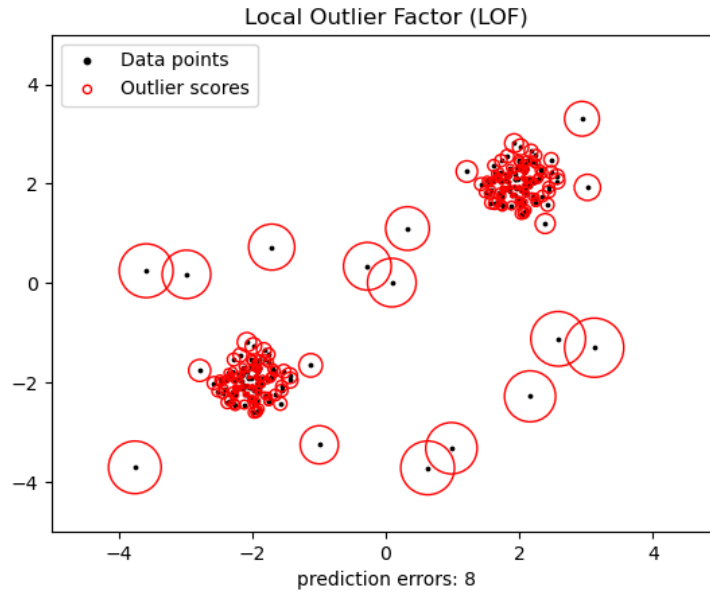


Figure 4: Local Outlier Factor detector

Detects anomaly based on Generalized Extreme Studentized Deviate (ESD) test (generalized Grubb's test). For predicting, the detector adds each value in the testing series to the set of normal values from training series independently, and performs generalized ESD test to this set (all normal values from training series, plus one value from testing series) to evaluate if this value of interest is an outlier. Assuming values follow an approximately normal distribution. **Parameters:** **alpha**, significance level.

`adtk.detector.AutoregressionAD`

Applying of a regressor to learn autoregressive property of the time series. For a linearly autoregressive time series, current value is a linear combination of several previous values. Violation of usual auto regressive behavior may indicate anomaly. **Parameters:** **n_steps**, number of previous values to use for learning model. **step_size**, length of a step. **c**, same as in PersistAD.

`sklearn.ensemble.IsolationForest`

The IsolationForest "isolates" observations by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature.

Multivariate

`adtk.detector.PcaAD`

Formally, Principal Component Analysis is a statistical technique for reducing the dimensionality of a dataset. In this context, performed for each multivariate time series (every time point is treated as a point in high-dimensional space), measures reconstruction error at every time point, and identifies a time point as anomalous when the reconstruction error is beyond anomalously large.

`adtk.detector.OutlierDetector - sklearn.neighbors.LocalOutlierFactor`

Derives a score (called local outlier factor) reflecting the degree of abnormality of the observations. By measuring the local density deviation of a given data point with respect to its neighbors, it detects the samples that have a substantially lower density than their neighbors. Figure 4 presents an example of the LOF algorithm.

`adtk.detector.MinClusterDetector - sklearn.cluster.KMeans`

Treating multivariate time series as independent points in a high-dimensional space, divides them into clusters, and identifies values in the smallest cluster as anomalous. Any clustering technique is supported, and on this instance, the classic KMeans algorithm was used: an iterative algorithm that tries to partition the dataset into K pre-defined distinct non-overlapping subgroups (clusters) where each data point belongs to only one group.

Each of the above algorithms has been implemented as a `.fit_detect()` method, in order to both train an anomaly detection model and provide an outlier detection and handling algorithm for each parameter. The previously mentioned Patras dataset is shown as an example of a workflow. For conductivity, figure 7 shows the process of step by step outlier labelling and removing. A boolean mask (series of boolean values that is applied on a series of data for selecting/labeling them) is created, and then is iterated: For every outlier (a boolean True), its value gets overwritten by the first occurrence of a normal value (a boolean False). For reference, the code is presented:

```
def outlierUPDATE(series, anomalies):
    cleanseries=series.copy()
    anomalies=anomalies.fillna(False)
    for i in range(len(anomalies)):
        if anomalies.iloc[i]:
            for d in range(i,len(anomalies)):
                if not anomalies.iloc[d]:
                    cleanseries.iloc[i]=cleanseries.iloc[d]
                    break
    return cleanseries
```

In this way, first step is a `GeneralESDTestAD()`, followed by an `IsolationForest()`, and finally an `AutoregressionAD()`. All three methods are fitted and saved for use with new data. By the same principle, on each parameter, the following result is obtained:

```
{'TMP': [<adtk.detector._detector_1d.PersistAD at 0x2932a1096d0>],

'CND': [<adtk.detector._detector_1d.GeneralizedESDTestAD at 0x29290e3b1c0>,
IsolationForest(bootstrap=True, contamination=0.1, n_estimators=200,random_state=0, warm_start=True),
<adtk.detector._detector_1d.AutoregressionAD at 0x2932a0ec7c0>],

'DO': [<adtk.detector._detector_hd.OutlierDetector at 0x2931dd6e6a0>,
<adtk.detector._detector_hd.OutlierDetector at 0x2930e17d3a0>,
<adtk.detector._detector_hd.PcaAD at 0x2932a155790>],

'PH': [<adtk.detector._detector_1d.PersistAD at 0x292bcb0ea30>,
<adtk.detector._detector_1d.AutoregressionAD at 0x292e94ef160>],

'ORP': [<adtk.detector._detector_1d.PersistAD at 0x292fd13d820>,
<adtk.detector._detector_hd.PcaAD at 0x292e94efd00>],

'NTU': [<adtk.detector._detector_1d.PersistAD at 0x292cfd7c4c0>]}
```

Notice how on dissolved oxygen, mainly multivariate detectors were applied, specifically for uncovering relationships between pH and temperature. Patras, Aigio, Actium and Preveza visualizations are shown in figures 8, 9, 10 and 11 respectively. The transformations used to review the timeseries are stored in a dataframe, accessible by location and parameter detector.

While being able to assess if a value is acceptable is a very powerful tool, it does not serve the full purpose of this study. A step forward is forecasting based on previously gathered information.

	TMP	CND	DO	PH	ORP	NTU
count	106591.000	106591.000	106591.000	106591.000	106591.000	106591.000
mean	20.216	50.109	6.508	10.613	308.327	0.673
std	6.241	5.713	2.504	3.280	175.753	2.462
min	8.120	19.780	0.000	6.130	-516.720	0.000
25%	14.260	48.450	5.060	8.330	118.160	0.000
50%	20.450	51.950	6.450	8.830	401.870	0.000
75%	26.340	53.450	7.790	14.420	431.750	0.400
max	29.710	59.820	17.630	20.910	485.760	287.820

Table 3: Preveza: raw data for previously described time range

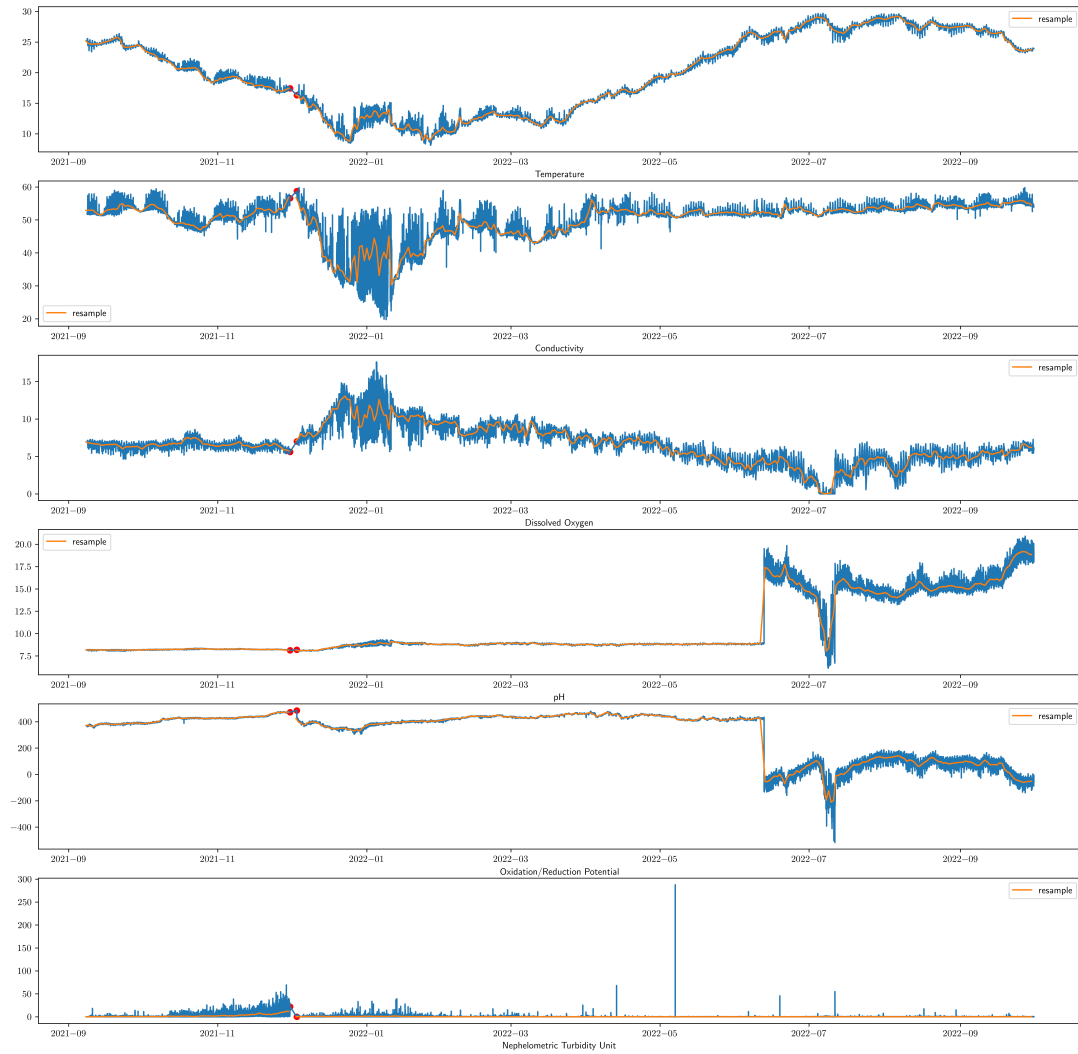


Figure 5: A comprehensive plot of all data gathered in Preveza. Red dots are timegaps > 1 day, orange line is a 1 day frequency resample.

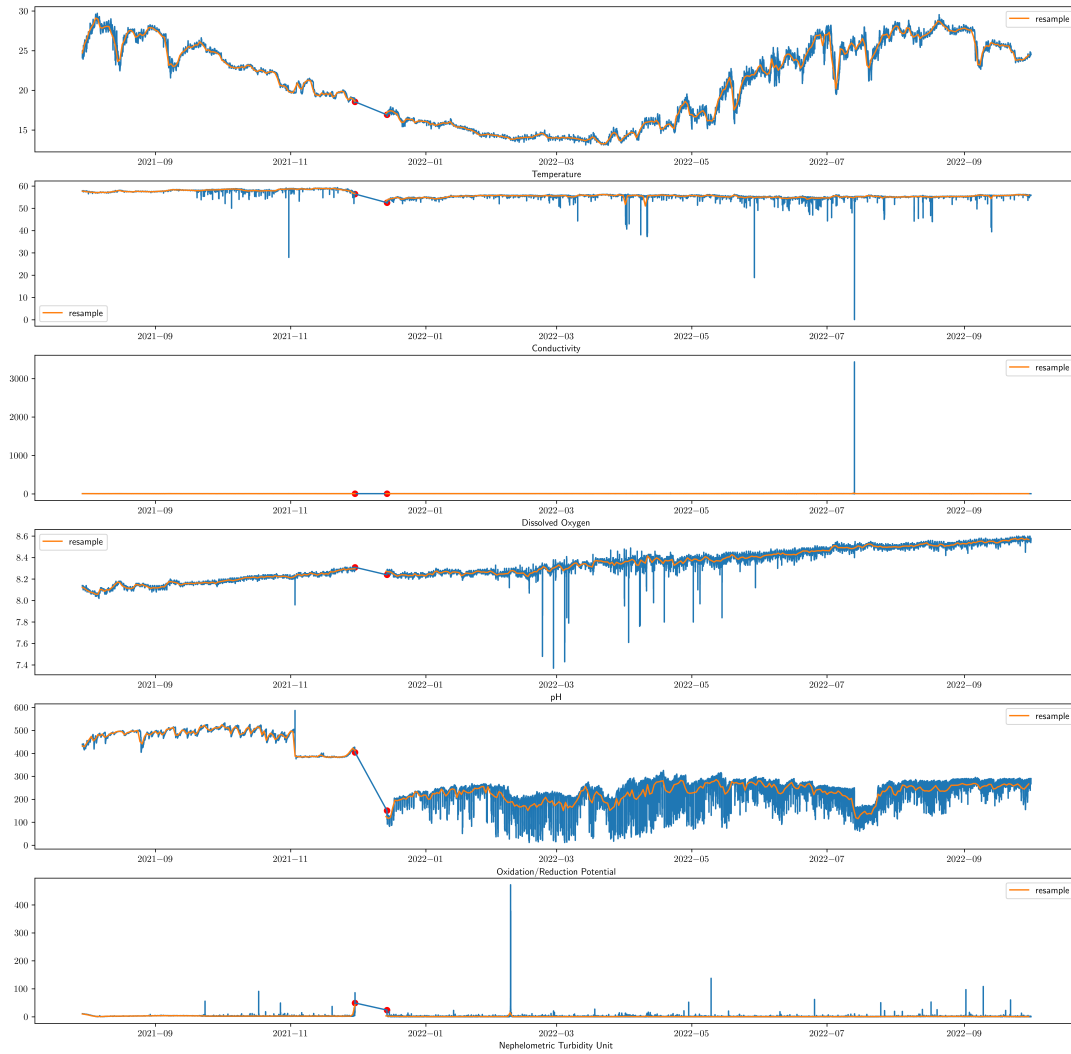


Figure 6: A comprehensive plot of all data gathered in Patras. Red dots are timegaps > 1 day, orange line is a 1 day frequency resample.

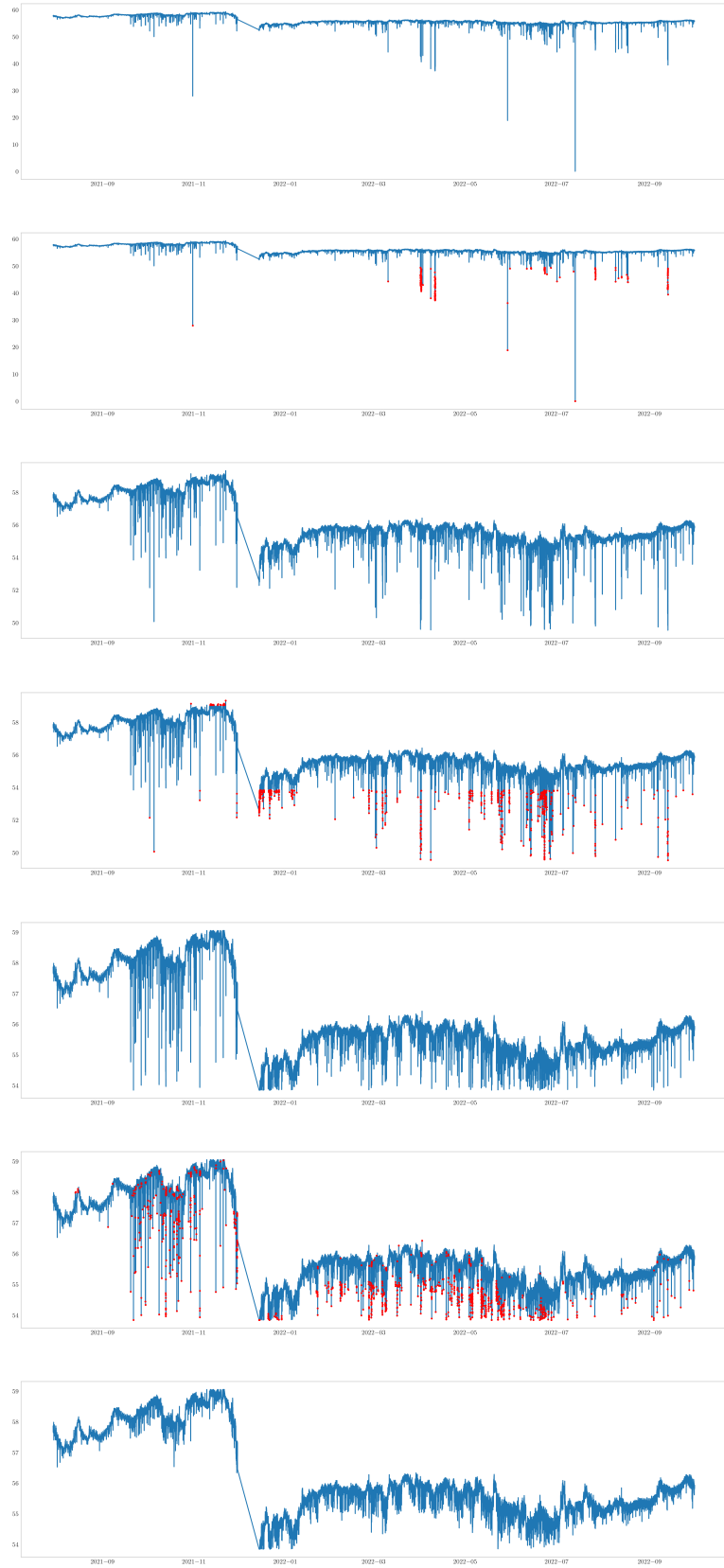


Figure 7: Conductivity measurements outlier/anomaly detection steps

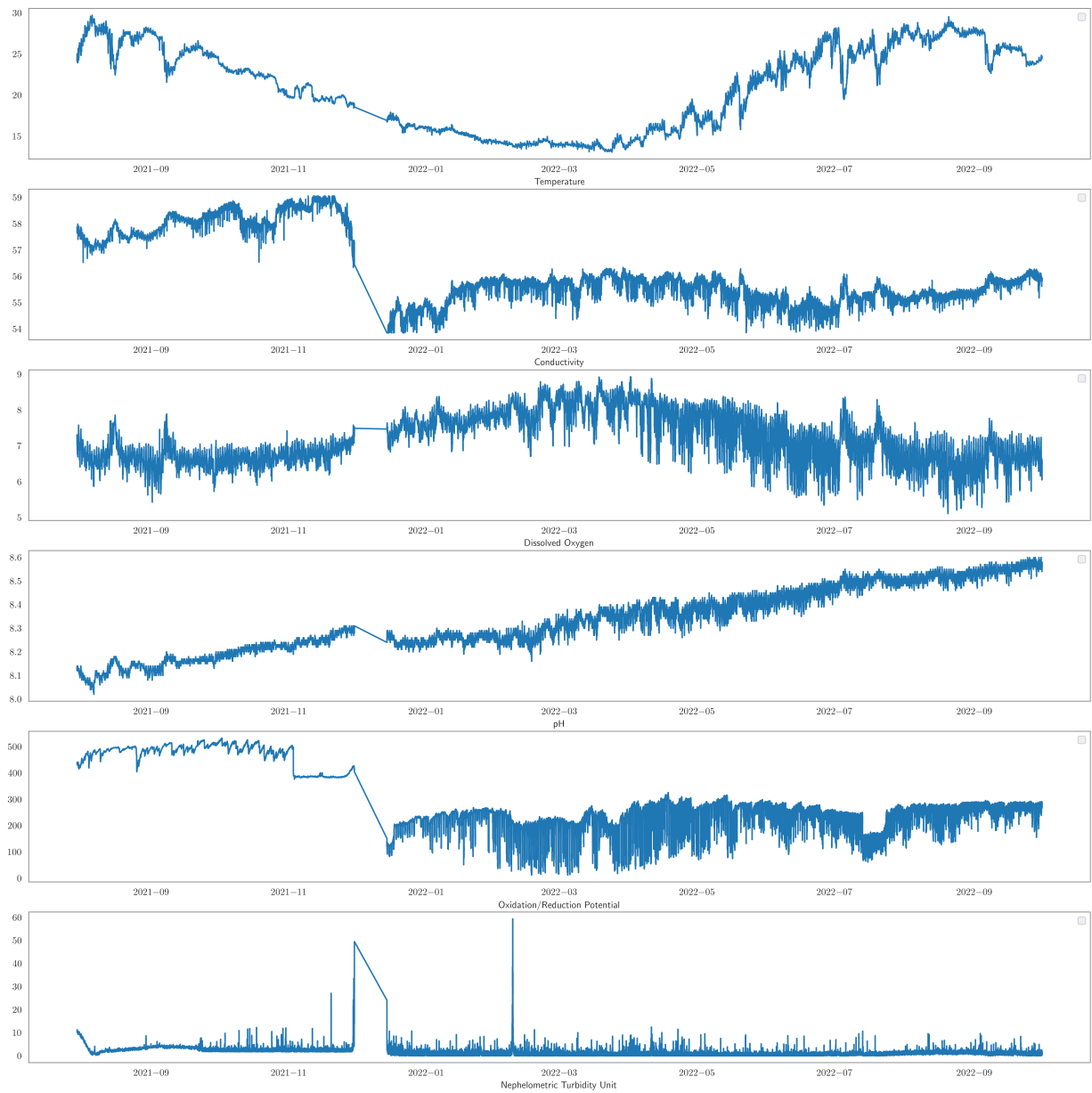


Figure 8: Reviewed and processed data from the **Patras** parameter set.

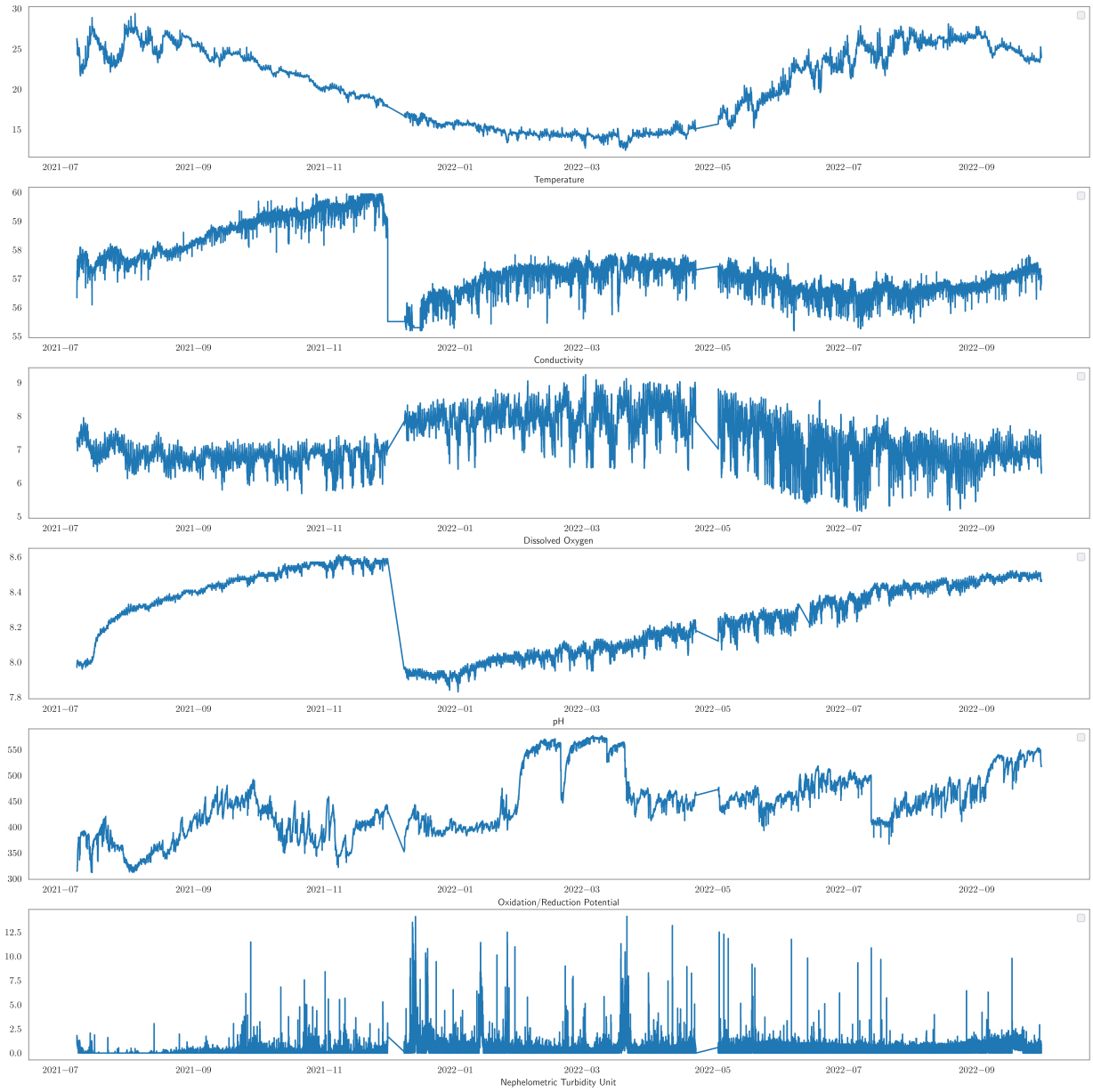


Figure 9: Reviewed and processed data from the **Aigio** parameter set.

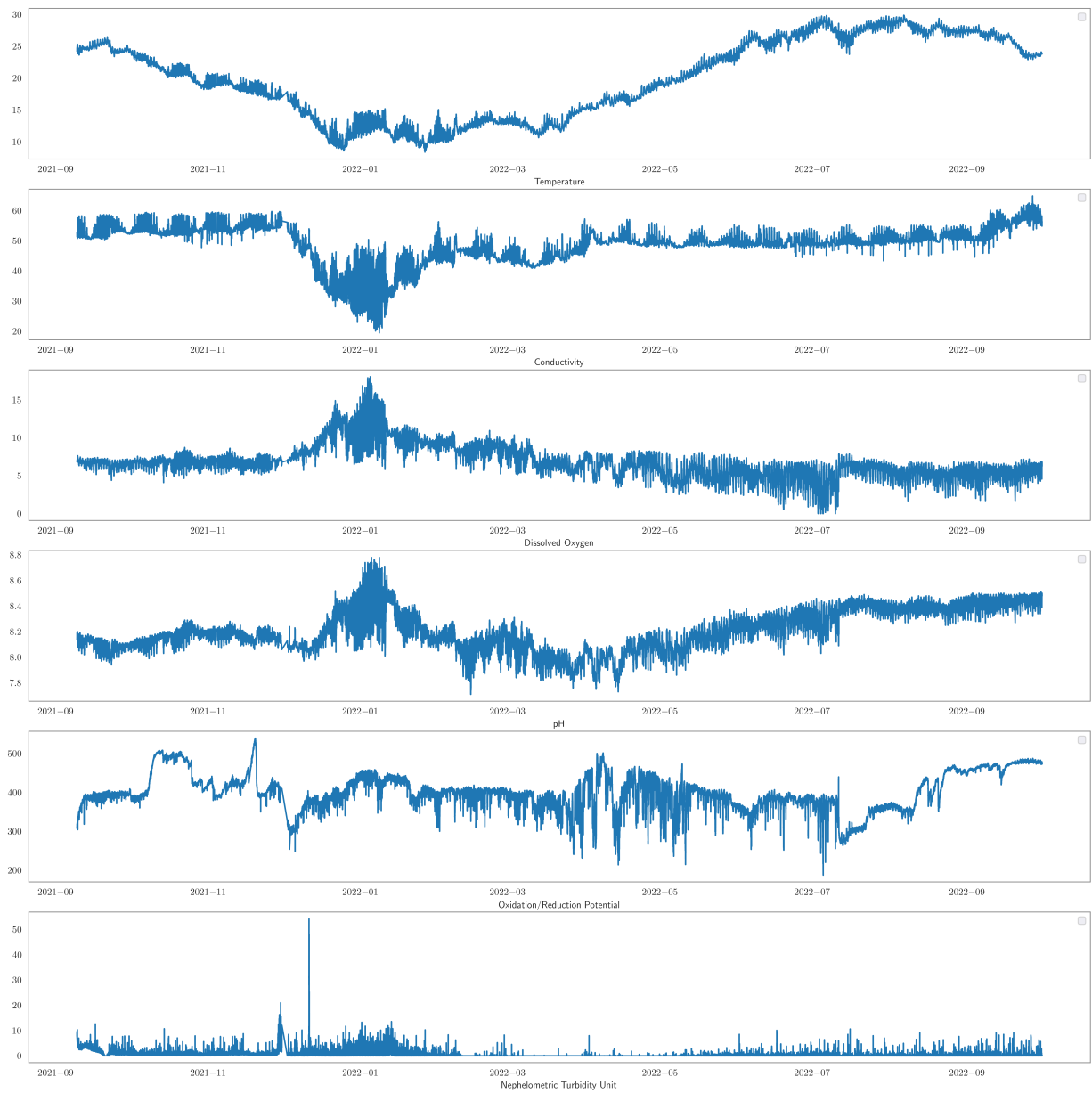


Figure 10: Reviewed and processed data from the **Actium** parameter set.

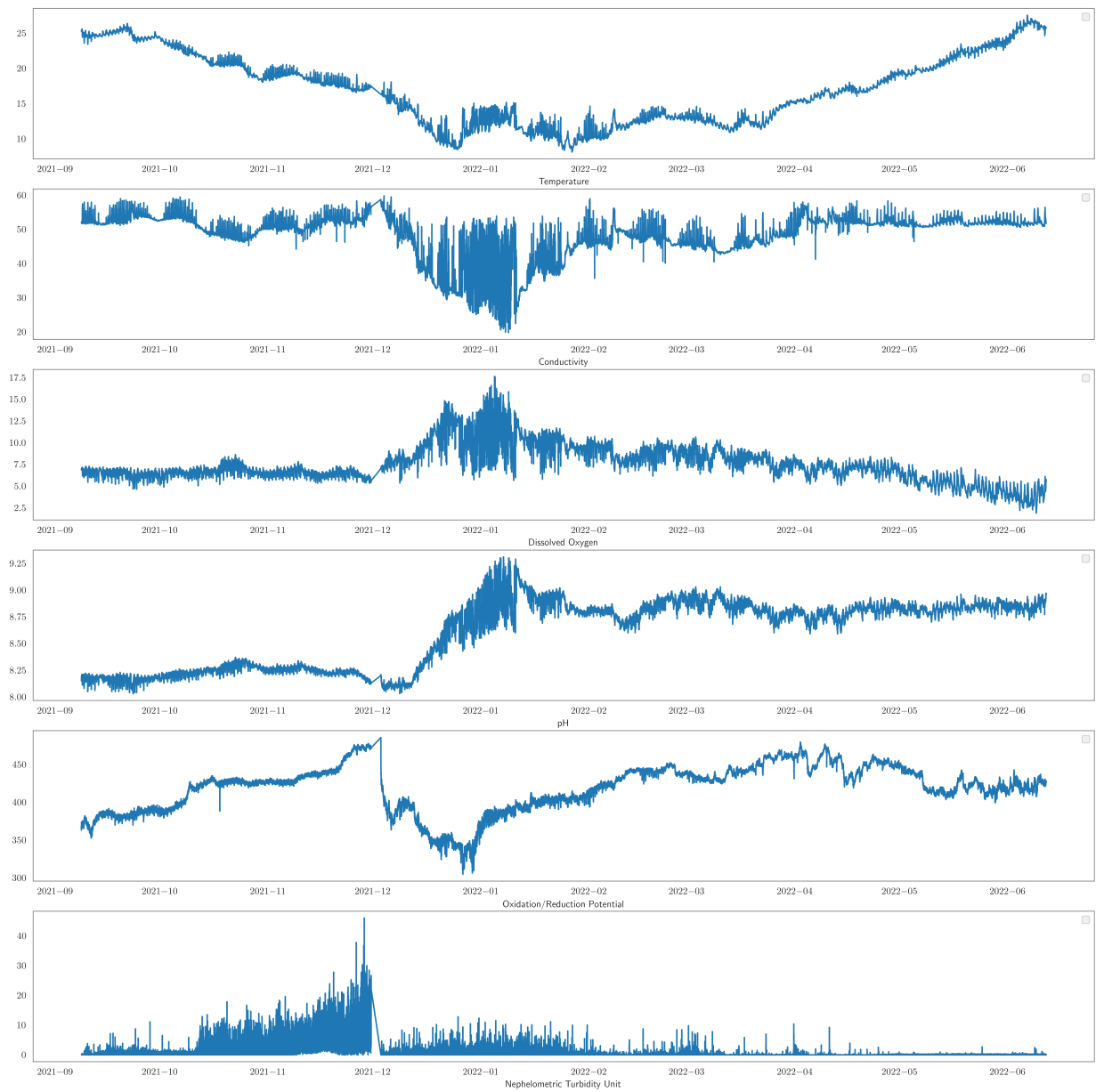


Figure 11: Reviewed and processed data from the **Preveza** parameter set.

4 Timeseries Forecasting

Predicting datasets is in no way a modern requirement, nor is the research surrounding methods of approximating the future. Fourier transforms used to model sea tides, moon eclipses, even simple calendars, in many ways humans have tried to prepare for the unknown. In the very limited scope of this article, a very basic prediction algorithm is used. The model is centered around the idea of combining an outlier detector with a timeseries predictor. Thus, in steps of prediction - detection - update the model is able to expect a certain turn of events, assess if they are anomalous, and in turn inform the decision maker.

`sktime.forecasting.ets.AutoETS`

Exponential Smoothing, used as a predictor method. Computes a weighted average over all parameter values in the input time series as its prediction. The weights are exponentially decreasing as time progresses, rather than the constant weights in simple moving average methods. The weights are dependent on a constant parameter: the smoothing parameter [15].

`ThetaForecaster()` is also used in cases where `AutoETS()` fails. To use the model, a `DataFrame` and a `PRESENT` datetime must be decided. Then, the past values range is set (in days), and the prediction horizon (usually 48 hours). For the example shown in figure 14, values from Actium dataset are explored, with a present being 2022-09-25 10:00, 20 days of training set and 48 hours of forecast. The reason for avoiding larger, or even complete dataset, training data is mainly the lack of seasonal data. If a larger past horizon is selected, yearly seasonality trends will be apparent to a human, but not to a model with no access to many years worth of information.

Complete Test To combine results of both anomaly labeling and forecasting, a skewed dataset is manually created as shown on figure 12. Deriving from Patras parameter value set, only TMP and DO is used. On DO, a `statsmodels.tsa.tsatools.add_trend` is applied, accounting for a quadratic trend meant to destabilize predictions. `PRESENT` is set as 2022-08-21 12:00. On the predicted values, a `Detectors.patra.DO[2]` detector is applied (multivariate DO and TMP), returning `True`(anomaly) for all values. Instead, when the same detector is tried on the control dataset (values of TMP and DO at the same datetimes, without the trend), no outliers are reported.

5 Limitations

To complete this brief overview on the developed code, limitations are discussed. The most important and time consuming issue, has been **data quality**. Missing, mislabeled, not conforming to specific formatting, all were problems that had to be dealt with. When the initial dataset is not "bulletproof", analytics and AI initiatives are time consuming, or even worthless to pursue.

Equally important, the area of "big data" does not necessarily refer to constant measuring, creating a barely manageable influx of data at neighboring time points. Especially when dealing with biochemical indicators and processes, **years** of information is needed to provide meaningful conclusions.

Even when appropriate quality and quantity of measurements is gathered, in a problem such as the one briefly discussed throughout this study, **labeling** is as important. Classification of measurements as anomalous, or even more complex, classification of measurements as a very specific type of pollution needs independent testing, to provide the necessary labeled data for training any model.

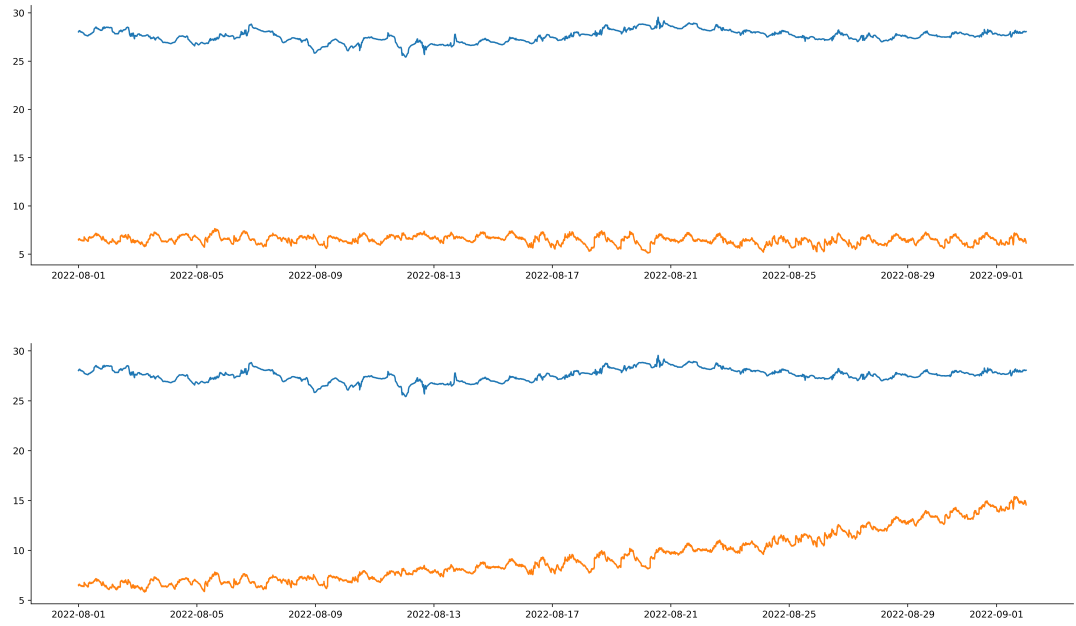


Figure 12: Subset of dataset, coupled with a quadratic trend for model testing

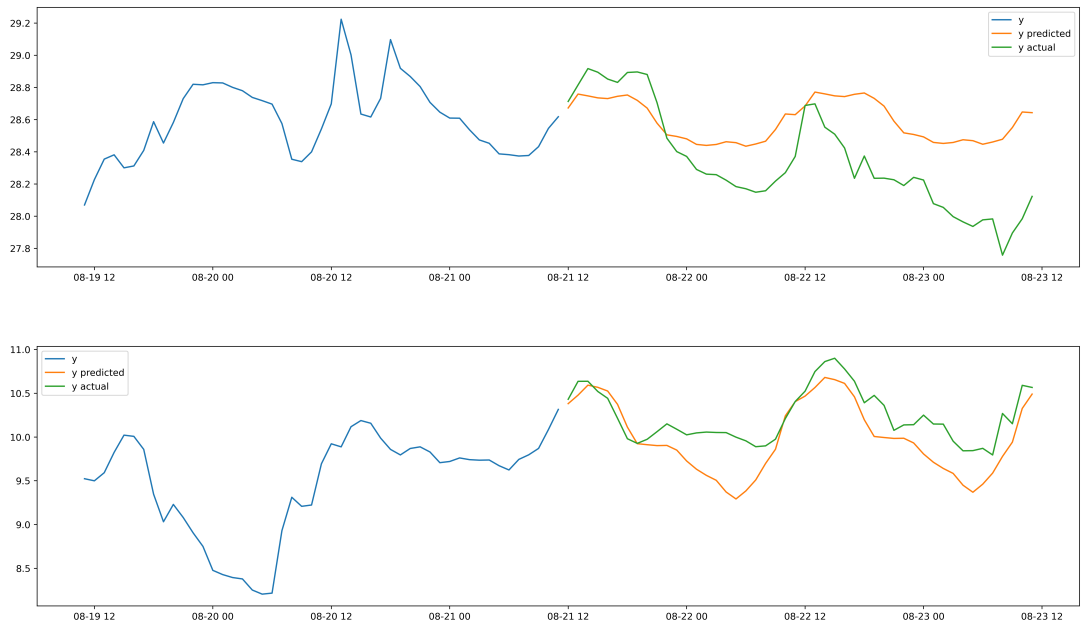


Figure 13: Predictions of TMP and DO for the trend dataset

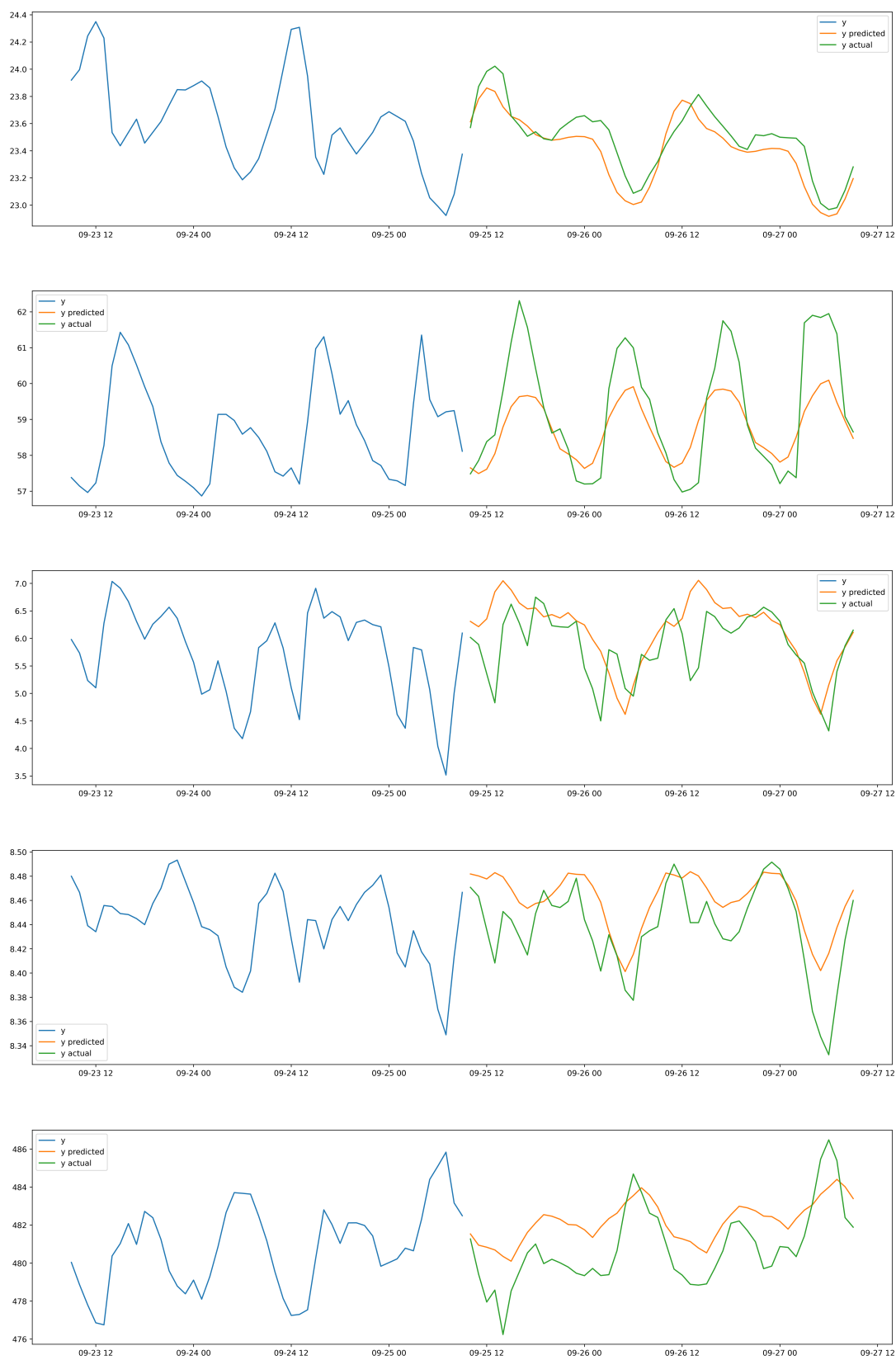


Figure 14: Prediction plots, from top to bottom: TMP, CND, DO, PH, ORP.

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